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ORIGINAL ARTICLE

# Social cognition differentiates phenocopy syndrome of behavioural variant frontotemporal dementia from behavioural variant frontotemporal dementia

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***Key words:*** *diagnostics, frontotemporal dementia, phenocopy FTD, social cognition.*

## Abstract

**Background:** Patients displaying clinical features of behavioural variant of frontotemporal dementia (bvFTD) but lacking both neuroimaging abnormalities and clinical progression are considered to represent the phenocopy syndrome of bvFTD (phFTD). Extensive clinical overlap between early phase bvFTD and phFTD hampers diagnostic distinction. We aimed to assess the diagnostic value of clinician-rated, self-reported and caregiver-reported symptoms for clinical distinction between phFTD and bvFTD.

**Methods:** There were 33 phFTD and 95 probable bvFTD patients included in the study (total $N = 128$). Clinician-rated, self-reported tests and caregiver-reported symptoms were compared between phFTD and bvFTD on social cognition, behaviour, mood and activities of daily living (ADL). Scores were compared between groups, followed by multiple logistic regression analysis, adjusted for age and sex. Receiver operating characteristic curves were plotted to assess diagnostic value.

**Results:** Using clinician-rated and self-reported tests, phFTD patients performed better on facial emotion recognition and reported more depressive symptoms. Caregiver-reported behavioural symptoms indicated higher behavioural and ADL impairment in phFTD compared to bvFTD. Facial emotion recognition provided highest diagnostic accuracy for distinction of phFTD from bvFTD (area under the curve (AUC) 0.813 95% CI 0.735–0.892, $P < 0.001$, sensitivity 81%, specificity 74%) followed by depressive symptoms (AUC 0.769 95% 0.674–0.864, $P < 0.001$ sensitivity 81%, specificity of 63%).

**Conclusion:** Social cognition tests are most suitable for distinction of phFTD from bvFTD. Caregiver-reported questionnaires and phFTD diagnosis seemed inversely correlated, showing more symptoms in phFTD. Further research is needed on phFTD aetiology and in caregivers taking into account disease burden to assess what explains this discrepancy between clinician-rated and caregiver-based tools.

## INTRODUCTION

The consensus criteria[1] of the behavioural variant of frontotemporal dementia (bvFTD) improved diagnostic accuracy by taking into account neuroimaging abnormalities, creating a level of probability, resulting in a 'possible', 'probable' or 'definite' bvFTD diagnosis. However, since 2006 a subgroup of patients was observed, presenting with the neuropsychiatric symptoms of bvFTD without typical frontotemporal neuroimaging abnormalities, thereby fulfilling the

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criteria of possible bvFTD, that do not convert to probable bvFTD during clinical follow-up. This phenomenon is now described as the phenocopy syndrome of FTD (phFTD).[2–8] Earlier research estimated that 7%–37% of the total group presenting with bvFTD-like symptoms represents this 'benign' or 'non-progressive' variant, later referred to as the phenocopy syndrome of bvFTD; one review even reported a range of 0–52%.[3,4] Many centres struggle with these patients, since the aetiology remains unclear. Previous studies hypothesised that phFTD might have a multifactorial non-degenerative cause such as a combination of subthreshold psychiatric diseases, relationship problems, recent life events and cluster C personality traits.[7,8] In contrast with these findings, functional imaging data have made others suggest phFTD should be considered as a prodromal disorder within the FTD spectrum.[9] Histopathological data on phFTD patients is sparse,[4] but three out of five described cases showed no neurodegenerative abnormalities. Nonetheless, the two remaining cases showed some signs of possible ubiquitin-positive inclusions.[4,6,10,11] Caution is also warranted with respect to carriers of the *C9ORF72* repeat expansion, that is associated with an extremely slow disease course with preserved neuroimaging.[12,13]

The clinical distinction of phFTD and bvFTD at presentation is challenging due to the great overlap in the nature of the behavioural symptoms, symptom duration at first presentation and severity as reported by spouses or family members. Moreover, functional and structural neuroimaging can be inconclusive, as it is often preserved in early disease stages of bvFTD.[2–5,14–16] Additionally, neuropsychological tests showed little discriminating value so far due to inconclusive or conflicting results including better, worse or similar performance in bvFTD compared to phFTD.[3,4,15] Accurate distinction of phFTD from bvFTD is essential to correctly inform patients, family and caregivers and to enable adequate prognosis predictions, both from clinical and management perspectives. The phenocopy syndrome of bvFTD is possibly treatable depending on its aetiology, contrary to bvFTD and thus in case of misdiagnosis, phFTD patients are not receiving optimal treatment. Moreover, improved diagnostic distinction contributes in correct bvFTD patient selection in future clinical trials.

As the aetiological hypotheses for phFTD are still broad and include (a combination of) psychological and psychiatric subthreshold disorders, domains that have showed discriminative value in discerning bvFTD from primary psychiatric disorders in previous research were examined. These include structured analysis of mood disorder symptoms, stereotypy, activities of daily living (ADL) functioning and social cognition tests assessing facial emotion recognition, that were all administered at baseline.[15–23] Additionally, since both phFTD and bvFTD patients usually lack adequate insight in symptoms and behaviour, the initial clinical assessment strongly depends on information of spouses or family members. Based on clinical experience, caregivers of phFTD patients often report equally severe symptoms, hampering diagnostic distinction with bvFTD. Therefore, we aimed to compare clinician-rated, self-reported and caregiver-reported behavioural symptoms, social cognition and ADL to assess their diagnostic value in differentiating phFTD from bvFTD.

## METHODS

### Participants

Subjects were participants of the Late Onset Frontal lobe (LOF) study and/or the Social Brain Project (SBP), recruited through the memory clinic of the Alzheimer Centre VUmc Amsterdam and the Old Age Psychiatry Department of GGZinGeest Amsterdam, the Netherlands, between April 2011 and June 2013 (LOF) and between 2016 and March 2020 (SBP). The LOF study and SBP are both multi-centre observational prospective naturalistic studies that aim to identify (prodromal) bvFTD and distinguish it from other diseases that cause predominant behavioural symptoms (full design has been previously described[24]). The SBP largely corresponds with the LOF study, but has a greater focus on social cognition by including a clinical battery that measures different levels of social cognition including perception, interpretation and regulation. Inclusion criteria of the SBP and/or LOF study include behavioural changes that gradually emerged between 40 and 70 years of age as demonstrated by a Frontal Behavioural Inventory (FBI)[25] score ≥ 11 and/or a Stereotypy Behavioural Inventory (SRI)[26] score ≥10. Patients were excluded in case of a Mini-Mental State Examination


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(MMSE) score ≤ 18. The clinical dementia rating scale (CDR) was used to indicate disease severity.

The study was approved by the Medical Ethical Committee of Amsterdam University Medical Centre (UMC) and all participants provided written and oral informed consent. The Medical Ethics Review Committee study ID is 2016.526 (A2019.409). The Medical Ethics Review Committee of VU University Medical Center is registered with the US Office for Human Research Protections (OHRP) as IRB00002991. The FWA number assigned to VU University Medical Center is FWA00017598.

If participants were unable to provide full consent due to cognitive impairment, this was obtained through the informal caregiver or legal representative. The study has been carried out in accordance with the Declaration of Helsinki.

## Diagnostic procedure

All patients underwent standardised extensive multidisciplinary diagnostic assessment, comprising of a clinical evaluation by a neurologist and a psychiatrist including the medical history, the informal caregiver-based history, neurological, psychiatric and physical examination, laboratory assessment, biomarker assessment in cerebrospinal fluid to exclude Alzheimer's disease pathology, neuropsychological assessment and neuroimaging (magnetic resonance imaging (MRI), and if MRI turned out to be inconclusive, additional [$^{18}$F]fluorodeoxyglucose-positron emission tomography ([$^{18}$F] FDG PET) scan) was performed. The diagnosis was determined in a multidisciplinary meeting including the neurologist, psychiatrist, radiologist and neuropsychologist and adjusted if appropriate during follow-up visits. In all patients, the diagnosis at the last follow-up visit was used for accurate patient group selection (at least 2 years follow-up). Diagnosis was stated using the diagnostic criteria for bvFTD[1] and the Diagnostic and Statistical Manual of Mental Disorders, (fourth edition from 2011 to 2013, fifth edition (DSM-V) 2013 and following) for psychiatric disorders.

### *BvFTD diagnosis*

All patients fulfilled diagnostic criteria for *probable* bvFTD.[1]

### *PhFTD diagnosis*

All cases with behavioural changes, interfering with daily life as reported by the informal caregiver (e.g. spouse, or family member) and leading to the referral, without any evidence of a neurodegenerative cause or a psychiatric condition explaining the behavioural changes at baseline screening and cases diagnosed with '*possible*' bvFTD[1] were examined in more detail. Medical records were extensively re-evaluated by WK, YP, SB, and PL to check for cases fulfilling the phenocopy syndrome of bvFTD: patients were included if they fulfilled the diagnostic criteria of *possible* bvFTD[1] in absence of significant frontotemporal abnormalities on neuroimaging and without clear functional decline over at least a 2-year follow-up period (range 2–9 years follow-up) as confirmed by clinical and neuropsychological follow-up evaluations.

In more detail: a standardised MRI acquisition protocol[27] for memory clinic patients was used and included a sagittal 3D heavily T1-weighted gradient-echo sequence with coronal reformats, a sagittal 3D T2-weighted fluid-attenuated inversion-recovery fast spin-echo with axial reformats, a transverse T2-weighted fast spin-echo, a transverse T2* susceptibility sequence, and diffusion-weighted imaging/echo planar imaging. Brain MRIs were assessed for signs of neurodegeneration, vascular damage, or other pathology that could explain the clinical symptoms.

Informal caregivers sometimes did report decline over the years, but this could not be confirmed by clinical and neuropsychological follow-up evaluations. The majority of bvFTD patients (91%) and all phFTD patients were tested for mutations in the *C9ORF72*, *MAPT*, or *GRN* genes. phFTD patients were excluded in case of a genetic mutation (*C9ORF72*, *MAPT*, or *GRN*) or when after second psychiatric evaluation a psychiatric condition was considered explanatory for the clinical syndrome.

## Social cognition, behavioural symptoms, mood and ADL

### *Clinician-rated and self-reported tests*

Social cognition was assessed using the Ekman 60 faces test.[28] The Ekman 60 faces test employs a selection of images sourced from the Ekman and Friesen collection of Pictures of Facial Affect,[29]

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acknowledged as the most extensively employed and validated set of images in facial expression research. Ten actors (six female, four male) from this collection were chosen, each depicting six basic emotions (happiness, sadness, disgust, fear, surprise, and anger). The Ekman 60 faces test serves as a tool for evaluating the recognition of facial expressions corresponding to basic emotions. The optimal test score, indicative of optimal performance, is 60, encompassing all six emotions, with a subset score of 10 for each individual basic emotion. Briefly, according to the Ekman 60 faces test manual,[28] healthy controls in the age range of 61–70 show a mean total test score of 49.41, with a cut-off at 41. Mood was assessed in a semi-structured interview for depressive symptoms using the Montgomery Åsberg Depression Rating Scale (MADRS)[30,39,40] (higher scores indicate more depressive symptoms).

### *Informal caregiver-reported symptoms*

Informal caregiver-rated questionnaires were used to assess behavioural symptoms using the Cambridge Behavioural Inventory (CBI)[31] (global cognitive functioning, ADL, behaviour, mood), the Frontal Behaviour Inventory (FBI)[25] (various behavioural symptoms), the Stereotypy Rating Inventory (SRI)[26] (compulsiveness/stereotype behaviour) and ADL using the Frontotemporal Dementia Rating Scale (FRS),[32] in which higher scores indicate worse performance/more symptoms in all questionnaires. Figure 1 displays an overview of clinician-rated and self-reported tests.

### Statistical analysis

All statistical analyses were performed in SPSS Statistics 26. Depending on the distribution, independent Student's *t*-test or Mann–Whitney *U*-test was used for continuous variables. For categorical variables $\chi^2$-test or Fisher's exact test was applied. Multiple logistic regression analyses were performed to compare test scores between groups and to compose predictive models for phFTD versus bvFTD while correcting for age and sex. Receiver operating characteristic (ROC) curves were plotted to assess diagnostic values.

## RESULTS

### Baseline characteristics

A cohort of $n = 128$ ($n = 33$ phFTD and $n = 95$ probable bvFTD) patients were included. The majority of phFTD and bvFTD patients was male, with the highest percentage in the phFTD group (male count phFTD 88%, bvFTD 61%, $P = 0.005$). There were no differences in age, CDR, education or symptom duration between phFTD and bvFTD (Table 1). phFTD patients had higher MMSE (phFTD: 27 ± 3.1, bvFTD: 25 ± 3.5; $P = 0.008$) and FAB scores (phFTD: 17 (14–18), bvFTD: 14 (11–17); $P < 0.001$) compared to bvFTD.

### Clinician-rated and self-reported tests – social cognition and mood

Compared to bvFTD patients, phFTD patients performed better on facial emotion recognition (Ekman 60 faces test mean ± SD: 41 ± 6.4 versus 31 ± 9.2, Exp(B) = 1,140 with a 95% CI of 1,073-1,212; $P < 0.001$, adjusted for age and sex). PhFTD patients reported more depressive symptoms (MADRS median (Q1–Q3): 11 (6–17) vs 4 (2–8), Exp(B) = 1,149 with a 95% CI of 1,071- 1,234; $P < 0.001$, adjusted for age and sex, (Table 2, Fig. 2). In all regression models, male sex was an effect modifier by decreasing the odds on phFTD diagnosis with predictors Ekman 60 faces test (odds 0.22) and MADRS (odds 0.21). ROC analysis of the Ekman 60 faces test comparing phFTD with bvFTD yielded an area under the curve (AUC) of 0.81, 95% CI 0,735–0,892, $P < 0.001$, sensitivity 81%, specificity 74% at cut-off 36.5. The ROC analysis of the MADRS comparing phFTD with bvFTD yielded an AUC of 0.77, 95% 0,674–0,864, $P < 0.001$, sensitivity 81%, specificity of 63% at cut-off 5.5 (Fig. 3).

### Informal caregiver-reported symptoms-behaviour and ADL

Using informal caregiver-rated questionnaires phFTD patients had equally severe compulsive symptoms, compared to bvFTD (SRI median (Q1–Q3): 8 (3–18) vs 9 (4–18), Exp(B) = 0,994 with a 95% CI of 0,954–1,035; $P = 0.767$). PhFTD patients had higher scores on impairment of global cognition, behavioural symptoms, mood and ADL measured with the CBI, compared to bvFTD patients (CBI mean ± SD: 62 ± 23.7


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| Test | Assesses: | Composition: | Completed by: | Cut-off score for bvFTD: |
|---|---|---|---|---|
| Ekman 60 Faces Test | Social cognition (facial emotion recognition) | 6 items (emotions), max. 10 points per item (max. 60 points) | Patient | - Age 41-60 years: <43 suspect for bvFTD<br>- Age 61-70 years: <41 suspect for bvFTD |
| MADRS | Depressive symptoms | (max. 60 points) | Patient | 0-8 = normal<br>9-18 = possible /light depression<br>19-26 = moderate depression<br>27-34 = severe depression<br>> 35 = very severe depression |
| SRI | Compulsiveness/stereotype behaviour | 5 items, frequency (4-point scale) times severity (3-point scale) | Informal caregiver | No cut-off score: increasing SRI score means more stereotypical behaviour |
| CBI | Global cognitive functioning, Activities of Daily Living (ADL), behaviour, mood | 45 items, frequency (4-point scale), (max. 180 points) | Informal caregiver | No cut-off score: increasing CBI score indicates increasing impairment |
| FBI | Multiple behavioural symptoms | 24 items, 4-point scale (none, mild, moderate, severe) | Informal caregiver | Cut-off score >30 suspect for bvFTD |
| FRS | Activities of Daily Living | 7 items, 3-point scale (always, sometimes, never) | Informal caregiver | Categories:<br>100-97%: very mild<br>96-80%: mild<br>79-41%: moderate<br>40-13%: severe<br>12-3%: very severe<br>2-0%: 'profound' |

**Figure 1** Overview of clinician-rated, self-reported and informal caregiver-reported tests. CBI, Cambridge Behavioural Inventory; FBI, Frontal Behaviour Inventory; FRS, Frontotemporal Dementia Rating Scale; MADRS, Montgomery Åsberg Depression Rating Scale; SRI, Stereotypy Rating Inventory.

versus 51 ± 22.9, Exp(B) = 1,022 with a 95% CI of 1,003–1,042; $P = 0.025$). Also, phFTD patients had more behavioural symptoms on the FBI, compared to bvFTD patients (FBI mean ± SD: 29 ± 10.3 versus 23.5 ± 10.1, Exp(B) = 1,054 with a 95% CI of 1,009-1,100; $P = 0.017$). PhFTD and bvFTD patients were equally moderately impaired in ADL (FRS mean ± SD: 48 ± 22.3 versus 56 ± 20.7, Exp(B) = 0,980 with a 95% CI of 0,961-1,000; $P = 0.051$) (Table 2, Fig. 2). In all regression models, male sex was an effect modifier by decreasing the odds on phFTD diagnosis with predictors SRI (odds 0.22), CBI (odds 0.20), FBI (odds 0.22), FRS (odds 0.18). For diagnostic distinction of phFTD from bvFTD, ROC analysis of the CBI and FBI yielded poor diagnostic accuracies: CBI AUC 0.641, 95% CI 0,527–0,756, $P = 0.022$; FBI AUC 0.641, 95% CI 0,526–0,756, $P = 0.022$ (Fig. 3).

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**Table 1** Demographics

| | All $N = 128$ | FTD $n = 95$ | phFTD $n = 33$ | *P*-value |
|---|---|---|---|---|
| Sex patient, male count (%) | 87 (68%) | 58 (61%) | 29 (88%) | 0.005[†] |
| Sex informant, male count (%) | 36 (29%) | 31 (34%) | 5 (15%) | 0.047[†] |
| Age (mean ± SD) | 65 ± 7.2 | 65 ± 7.3 | 66 ± 7.2 | 0.574[‡] |
| CDR (mean ± SD) | 1 ± 1 | 1 ± 0.6 | 1 ± 0.7 | 0.362[‡] |
| MMSE (mean ± SD) | 25 ± 3.5 | 25 ± 3.5 | 27 ± 3.1 | 0.008[‡] |
| FAB median [Q1–Q3] | 15 [12–17] | 14 [11–17] | 17 [14–18] | <0.001[§] |
| Education, years (mean ± SD) | 10.8 ± 2.9 | 10 ± 2.9 | 10 ± 2.8 | 0.372[‡] |
| Disease duration, years (mean ± SD) | 4.4 ± 3.1 | 4.6 ± 3.4 | 3.8 ± 2.1 | 0.112[‡] |

[†] Fisher exact test. [‡] Independent *t*-test. [§] Mann–Whitney *U*-test. Abbreviations: CDR, Clinical Dementia Rating Scale; FAB, Frontal Assessment Battery; FTD, frontotemporal dementia; MMSE, Mini-Mental State Exam; phFTD, phenocopy syndrome of FTD.

**Table 2** Social cognition, mood, behavioural and ADL questionnaires

| | All $n = 128$ | FTD $n = 95$ | phFTD $n = 33$ | *P*-value |
|---|---|---|---|---|
| Assessed in patient | | | | |
| Ekman 60 faces test (mean ± SD) | | 31 ± *9.2* | 41 ± 6.4 | <0.001[†] |
| MADRS median [Q1–Q3] | | 4 [2–8] | 11 [6–17] | <0.001[‡] |
| Informant-rated behavioural questionnaires | | | | |
| SRI median [Q1–Q3] | | 9 [4–18] | 8 [3–18] | 0.834[‡] |
| CBI (mean ± SD) | | 51 ± *22.9* | 62 ± 23.7 | 0.025[†] |
| FBI (mean ± SD) | | 23.5 ± 10.1 | 29 ± 10.3 | 0.01[†] |
| Informant-rated questionnaire of ADL | | | | |
| FRS (mean ± SD) | | 56 ± 20.7 | 48 ± 22.3 | 0.078[†] |

[†] Independent *t*-test. [‡] Mann–Whitney *U*-test. Abbreviations: ADL, activities of daily living; CBI, Cambridge Behavioural Inventory; FBI, Frontal Behaviour Inventory; FTD, frontotemporal dementia; FRS, Frontotemporal Dementia Rating Scale; MADRS, Montgomery Åsberg Depression Rating scale; phFTD, phenocopy syndrome of FTD.

## DISCUSSION

We present a study on clinical differentiation between phFTD and bvFTD patients with a design enabling comparison of informal caregiver-reported symptoms, clinician-rated and self-reported tests. Our study reveals that when using the clinician-rated Ekman 60 faces test, patients with phFTD demonstrate borderline impairment in social cognition according to the test norms of healthy controls,[28] but perform evidently better compared to bvFTD patients. Second, phFTD report more depressive symptoms using a combined clinician-rated and self-reported scale (MADRS). Facial emotion recognition (Ekman 60 faces test) provided the highest diagnostic potential to discriminate phFTD from bvFTD. Moreover, tests based on informal caregiver-reported information and diagnosis were inversely correlated by showing more behavioural symptoms in phFTD. Also, we found a clear male predominance in phFTD patients corresponding to previous studies[4] which remains unexplained. This is essential knowledge for clinical practice since the bvFTD differential diagnostic process depends heavily on the informant.

Our finding of spared social cognition in phFTD compared to bvFTD, is in line with other studies showing that early impairment of social cognition is a key hallmark of bvFTD, therefore aligning with the expectation of lower performances on social cognition testing in the bvFTD sample.[23,32,33] Although data on social cognition in phFTD patients are sparse, in a previous longitudinal study worse emotion recognition and sarcasm detection was found to differentiate progressive bvFTD from non-progressive bvFTD, thus reinforcing our findings.[33]

We found equal impairment in informal caregiver-reported ADL in phFTD and bvFTD, showing both 'moderate severity', which might reflect high caregiver burden in both conditions. Interestingly, in a previous study [15]no differences were found between phenocopy and bvFTD patients in informal caregiver-reported ADL functioning. This study [15]shows similar severity in ADL interference in both groups, whereas



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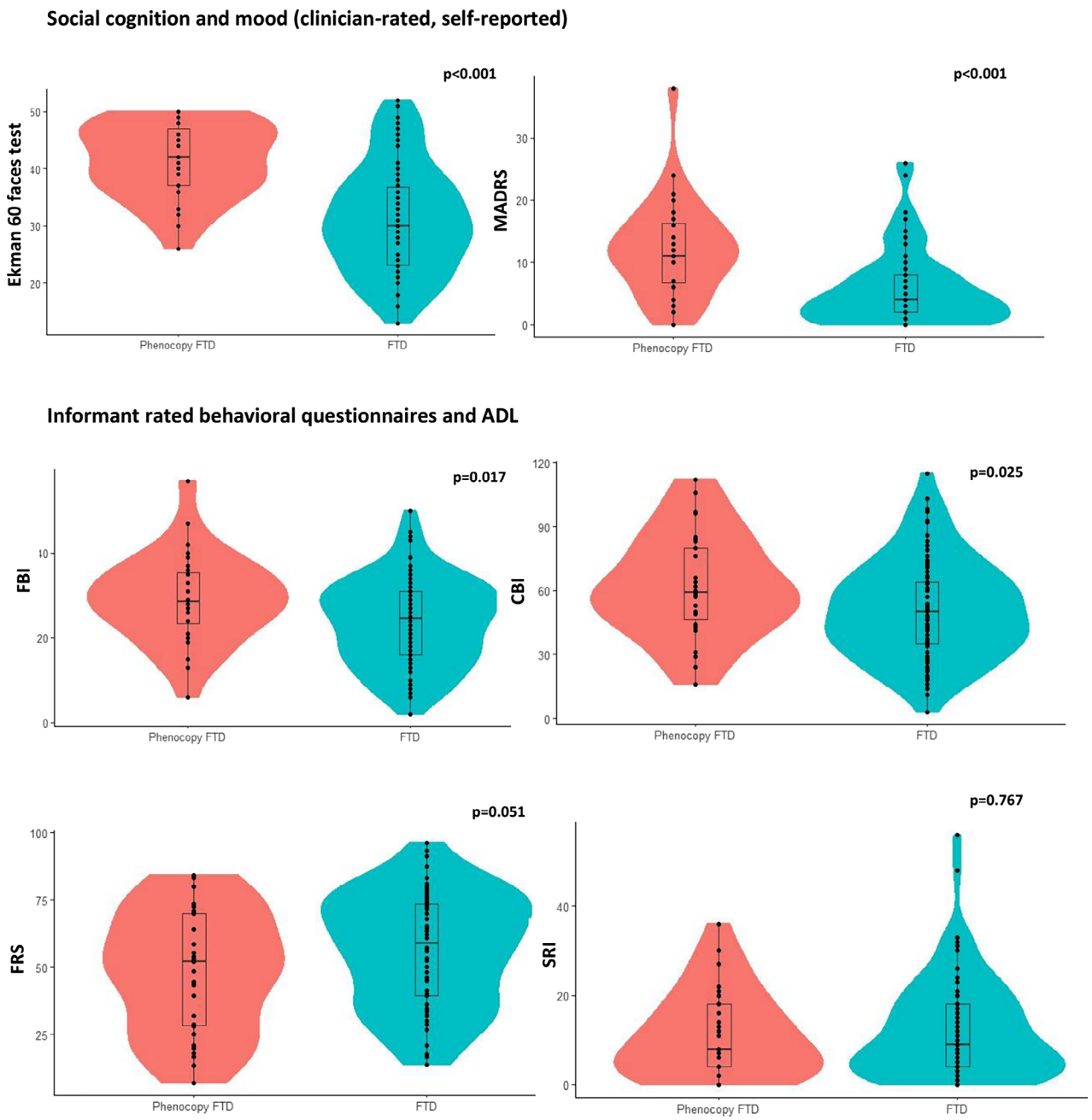


**Figure 2** Differences in social cognition, mood, behaviour and activities of daily living (ADL). CBI, Cambridge Behavioural Inventory; FBI, Frontal Behaviour Inventory; FTD, frontotemporal dementia; FRS, Frontotemporal Dementia Rating Scale; MADRS, Montgomery Åsberg Depression Rating Scale. Data represents group differences in total test scores using logistic regression analysis, adjusted for age and sex.

a clear distinction was found on the patient performance-based measure (social cognition), in which phFTD patients showed a significantly better performance, in line with our findings. Our findings of equally high or higher informal caregiver-reported behavioural symptoms and ADL impairment in phFTD compared to bvFTD, along with clinical observations indicating discrepancies between caregiver-reported and clinically observed behavioural symptoms, may indicate informal caregiver overreporting.[15] This might be partly driven by a combination of intrinsic parts of the underlying problem (e.g. marital problems), informal caregiver burden and uncertainty about the prognosis of phFTD due to a lack of

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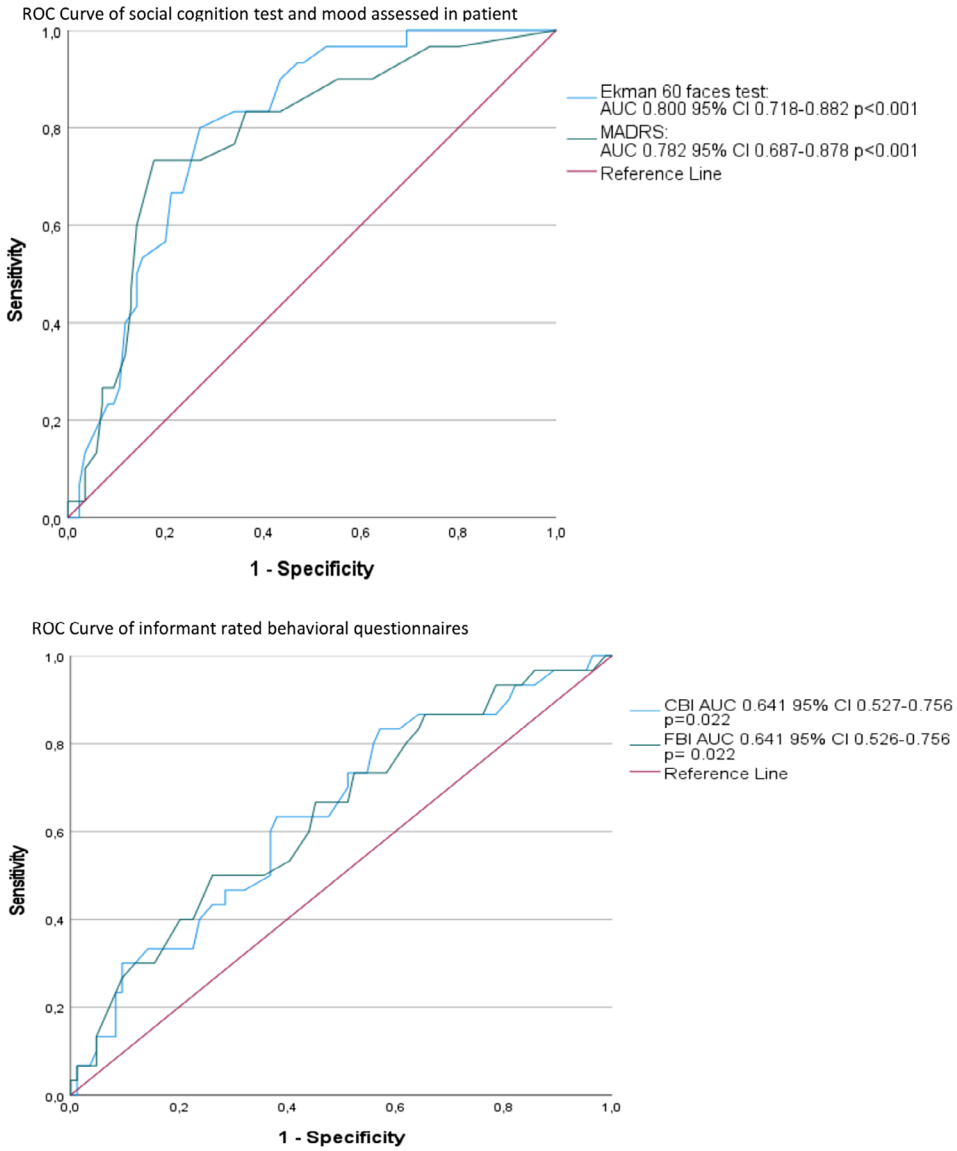


**Figure 3** Receiver operating characteristic (ROC) analysis of phenocopy FTD versus bvFTD. CBI, Cambridge Behavioural Inventory; FBI, Frontal Behaviour Inventory; bvFTD, behavioural variant of frontotemporal dementia. Data represent diagnostic accuracy comparing phenocopy FTD versus bvFTD, using patient based tests (social cognition, mood) and informant-rated behavioural questionnaires (CBI, SRI, FBI). State variable = phenocopy FTD (case) versus FTD (control). In both ROC analyses, higher scores indicate positive test.

aetiological diagnosis, but more research is needed to further explain these discrepancies. In case of overreporting by the informal caregiver or marital problems, these findings might give directions for therapy and management as it represents the possible reversible nature of phFTD.

Although our findings of the usefulness of the Ekman 60 faces test and the MADRS in discerning phFTD from bvFTD, this is in itself no proof of any particular aetiology in phFTD, but it does correspond with earlier findings in a study differentiating bvFTD from primary psychiatric disorders. As in phFTD, the patients with primary psychiatric disorders showed relatively spared social cognition and more depressive symptoms.[21,34] Nevertheless, the MADRS total scores were still relatively low in both groups (bvFTD: 4 (2–8), phFTD: 11 (6–17)), reflecting either no depression (0–8) or a possible/light depression (9–18). In our clinical experience, similar to bvFTD patients, anosognosia in phFTD is often accompanied by a lack of disease burden, which might explain the relatively low scores on the self-reported MADRS.

The above described findings and context could be in line with the previously suggested hypothesis of phFTD being a cluster or sum of multiple subthreshold psychiatric disorders. Other hypotheses on phFTD aetiology might include a very slow


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progressive neurological disorder of unknown origin, a factitious disorder (by proxy) or a disease entity within the spectrum of functional neurological syndromes. Inconsistencies within certain domains at neuropsychological examination and inconsistencies between clinician observations and reported symptoms may suggest the presence of functional neurological symptoms and could represent a significant aspect of the underlying aetiology. In phFTD, several similarities with functional cognitive disorder have also been observed, including slow information processing, increased distractibility, subthreshold anxiety complaints or decreased mood, a certain hyperfocus of the patient or spouse on the symptom complexes, recent life events and difficulties with mentalisation or metacognition.[35] More research is needed to explore this hypothesis and quantify the observations.

Interestingly, a recent study reporting normative data on facial emotion recognition in an Italian cognitively healthy population showed significant effects of demographic variables, in which females showed higher scores on the Ekman 60 faces test compared to males.[36] Combining this with the results from another study[8] in which phFTD patients had a history with relatively more cluster C personality traits or autism spectrum traits, another route of thinking about the phFTD aetiology might be some form of 'growing into deficits' in these patients, in which previous relative vulnerable qualities on social cognition or coping capabilities get compromised by life events, normal ageing or other influences.

A limitation of our study is the absence of autopsy-confirmed cases of phFTD. Due to the lack of disease progression, neuropathologically assessed phFTD cases are sparse. Future histopathological assessment on a larger scale is essential for elucidating aetiological factors of phFTD. Further, our study has not included neurofilament light chain (NfL) analyses as these data were not available, but recent research had shown that plasma NfL appears to differentiate FTD syndromes and mimics like phFTD, thus adding NfL analysis is likely to further improve diagnostic differentiation.[37] Also, patient groups in all studies are small, thus pooling of international data and preferably following phFTD patients prospectively is recommended for future research.

In the Netherlands, we observe an intensive search for an explanatory diagnosis throughout the healthcare system by phFTD patients and their families, alongside with sometimes second, third or even fourth opinions due to persistence of family members in their perceived recognition of a 'true' bvFTD. This is understandable and probably due to the lack of appropriate guidance and treatment available at the time, but increases the risk for iatrogenic damage such as misdiagnosis, hospitalisation and inappropriate (pharmacological) treatments. Clinicians should be aware of this risk. More knowledge on differentiation and aetiology, but particularly on management and prognosis of phFTD is needed.

This leads to practical recommendations in patients fulfilling possible bvFTD criteria, but lacking progression and imaging abnormalities. We advise testing facial emotion recognition and assessing mood in addition to the regular work-up. The combination of NfL and relevant clinical parameters including the Ekman 60 faces test and a mood assessment might be an even stronger clinical discriminator, but this should be examined in more detail. Further, we would advise performing a second informant-based history when possible, preferably with an informant suggested by the patient and outside the direct social circle of the first informant, as tests based on informal caregiver-reported information were not indicative and seemed inversely correlated, showing more symptoms in phFTD. Future research is needed to assess what explains this discrepancy between clinician-rated, self-reported and informal caregiver-based tools. In addition, we recommend a careful psychiatric evaluation, preferably taking multiple sessions to obtain biographical information, beyond simply applying uniform classifications. A clear single psychiatric disorder explaining the clinical features should be considered as exclusion criteria for phFTD. A *C9ORF72* repeat expansion needs to be excluded and at least a 2-year follow-up to evaluate (the absence of) progression is required. The latter is supported by another study showing that a bvFTD diagnosis remains stable enough to conclude that 2 years is sufficient to say if a patient with late-life behavioural disorder has bvFTD.[38] Management should encompass a personalised approach targeting mood, considering cognitive abilities, functional remediation and in some cases therapy focusing on acceptance seems appropriate, ideally including spouse and family members.

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## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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